Antiderivatives and Indefinite Integrals

Section 1: The Power Rule for Integration

1 Evaluate $\int (6x^3 - 4x + 5)\,dx$.

a. $\frac{3}{2}x^4 - 2x^2 + 5x + C$

b. $2x^4 - 2x^2 + 5x + C$

c. $18x^2 - 4 + C$

d. $\frac{3}{2}x^4 - 4x^2 + 5x + C$

2 Find $\int \left(x^{2/3} + \frac{1}{x^3}\right) dx$.

a. $\frac{3}{5}x^{5/3} - \frac{1}{2x^2} + C$

b. $\frac{5}{3}x^{5/3} - \frac{1}{4x^4} + C$

c. $\frac{3}{5}x^{5/3} + \frac{1}{2x^2} + C$

d. $\frac{2}{5}x^{5/3} - \frac{2}{x^2} + C$

3 Evaluate $\int \frac{3x^4 - 2x}{x^2}\,dx$.

a. $\frac{3}{3}x^3 - 2\ln|x| + C = x^3 - 2\ln|x| + C$

b. $x^3 - \frac{2}{x} + C$

c. $3x^3 - 2x + C$

d. $x^3 - 2x^{-1} + C$

Section 2: Trigonometric and Exponential Antiderivatives

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4 Evaluate $\int (3\sin x + 4\cos x)\,dx$.

a. $-3\cos x + 4\sin x + C$

b. $3\cos x - 4\sin x + C$

c. $3\cos x + 4\sin x + C$

d. $-3\cos x - 4\sin x + C$

5 Evaluate $\int (5e^x + \sec^2 x)\,dx$.

a. $5e^x + \tan x + C$

b. $5e^x - \tan x + C$

c. $5e^x + \sec x \tan x + C$

d. $e^x + \tan x + C$

6 Find $\int (2^x + \csc x \cot x)\,dx$.

a. $\frac{2^x}{\ln 2} - \csc x + C$

b. $2^x \ln 2 - \csc x + C$

c. $\frac{2^x}{\ln 2} + \csc x + C$

d. $2^x - \cot x + C$

Section 3: Initial Value Problems

7 Find $f(x)$ given $f'(x) = 4x - 3$ and $f(1) = 6$.

a. $f(x) = 2x^2 - 3x + 7$

b. $f(x) = 2x^2 - 3x + 5$

c. $f(x) = 4x^2 - 3x + 5$

d. $f(x) = 2x^2 - 3x$

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8 A curve satisfies $\frac{dy}{dx} = 3x^2 + 2x$ and passes through $(2, 10)$. Find its equation.

a. $y = x^3 + x^2 - 2$

b. $y = x^3 + x^2 + 2$

c. $y = 3x^3 + 2x^2 - 22$

d. $y = x^3 + x^2$

9 Given $f''(x) = 6x$, $f'(0) = 4$, and $f(0) = 1$, find $f(x)$.

a. $f(x) = x^3 + 4x + 1$

b. $f(x) = 3x^3 + 4x + 1$

c. $f(x) = x^3 + 2x + 1$

d. $f(x) = x^3 + 4$

Section 4: Rectilinear Motion and Acceleration

10 A particle moves with acceleration $a(t) = 6t - 4$. If its initial velocity $v(0) = 5$, find $v(t)$.

a. $v(t) = 3t^2 - 4t + 5$

b. $v(t) = 6t^2 - 4t + 5$

c. $v(t) = 3t^2 - 4t$

d. $v(t) = 3t^2 + 4t + 5$

11 If a ball is thrown upward with $v(0) = 48$ ft/s and $a(t) = -32$ ft/s^2 from ground level $s(0) = 0$, find its maximum height.

a. 36 ft

b. 48 ft

c. 64 ft

d. 32 ft

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12 A car decelerates at a constant $a = -4\ \text{m/s}^2$ from $v_0 = 24\ \text{m/s}$. How far does it travel before coming to a stop?

a. $72\ \text{m}$

b. $48\ \text{m}$

c. $96\ \text{m}$

d. $36\ \text{m}$

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